Shrey Patel

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Education

Bachelor of Science, Computer Science **May 2025**
Virginia Tech *GPA: 3.86*
Relevant Courses: Data Structures and Algorithms, Operating Systems, Computer Architecture, Database Management Systems, Cloud Software Development, Intro to AI, Machine Learning, Urban Visualization

Technical Skills

Languages: Java, Go, Python, TypeScript/JavaScript, C/C++, C#, SQL, HTML, CSS
Frameworks & Libraries: Spring, .NET, Flask, React, Node.js, LangChain, JUnit, Cypress, Jest
Messaging & Databases: RabbitMQ, Kafka, PostgreSQL, MongoDB, Redis
Cloud/DevOps: AWS, Docker, Kubernetes, Jenkins, Terraform, Git

Experience

Full-Stack Software Engineer **Oct 2025 – Present**
Integration Innovation *Huntsville, AL*

- Designed and shipped key architectural components of a web-based UAS collaborative ground control client (GCS), advancing a prototype toward a production-ready system.
- Developed testing frameworks and CI/CD pipelines for the GCS client, supporting integration, deployment, and collaboration with vendors and program stakeholders.
- Engineered reusable interaction and rendering workflows for complex geospatial interfaces, improving consistency across mission-planning tools.
- Contributed to fielded autonomy software for uncrewed aerial and ground platforms as part of a cross-functional team, supporting on-site system integration events.

Full-Stack Software Engineer Intern **Aug 2024 – Feb 2025**
Peraton *Blacksburg, VA*

- Engineered high-throughput Java microservices to orchestrate real-time satellite telemetry streams across 5 distributed ground stations, ensuring zero data loss for concurrent flight controllers.
- Reduced P99 message latency by 40% under heavy loads by isolating workloads into subsystem-specific RabbitMQ channels.
- Partnered with a cross-functional Agile team to deliver new functionality, inform stakeholders, and track KPIs via Jira and Confluence.

Full-Stack Software Engineer Intern **May 2024 – Aug 2024**
CACI *Reston, VA*

- Built CI/CD pipelines utilizing Terraform to automate AWS resource creation, detect configuration changes, post PR comments with Terraform plans, and enforce approval steps.
- Optimized data validation queries across 8M+ records using composite indexes, reducing query latency by 80%.

Full-Stack Software Engineer Intern **May 2023 – Aug 2023**
Ventera *Reston, VA*

- Scaled Node.js APIs by transitioning high-traffic endpoints to a Redis cache-aside architecture, cutting Time-to-First-Byte (TTFB) by 90% and increasing throughput 3x.
- Identified over-provisioned Kubernetes workloads by developing a performance testing suite integrated with Jenkins, reducing CPU usage by 25% and memory/pod counts by 15%.

Projects

Jobhub.cs.vt.edu | *React, Flask, PostgreSQL, Docker, Kubernetes* **Capstone Project**

- Designed and shipped an university industry-partnership portal for Virginia Tech's CS department as primary backend developer on a 5-person team, serving 100+ industry partners.

Personal Portfolio Website | *NextJS, OpenAI, LangChain, Vercel AI SDK, Redis* **Aug 2025**

- Implemented a RAG pipeline using LangChain, embedding personal context into a vector store and routing semantic queries through GPT-5 via Vercel's AI SDK.
- Added a serverless Redis caching layer for repeated queries, reducing inference latency and cutting operational cost.